

Will a Customer Accept the Coupon?

Practical Application 5.1 | UC Berkeley ML & AI Program

The goal of this analysis is to use visualizations and probability distributions to understand the difference between customers who accepted a driving coupon versus those who did not. The dataset comes from a survey on Amazon Mechanical Turk and includes information about the driving scenario, passenger, weather, time, and coupon type.

1. Setup and Data Loading

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# basic plot styling
sns.set_theme(style='whitegrid')
%matplotlib inline
```

```
In [2]: df = pd.read_csv('coupons.csv')
print(df.shape)
df.head()
```

(12684, 26)

```
Out[2]:
```

	destination	passanger	weather	temperature	time	coupon	expiration	gender
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Femal
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Femal
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Femal
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Femal
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Femal

5 rows × 26 columns



2. Data Overview and Cleaning

```
In [3]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12684 entries, 0 to 12683
Data columns (total 26 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   destination                           12684 non-null  object
1   passanger                             12684 non-null  object
2   weather                               12684 non-null  object
3   temperature                           12684 non-null  int64
4   time                                  12684 non-null  object
5   coupon                                12684 non-null  object
6   expiration                             12684 non-null  object
7   gender                                12684 non-null  object
8   age                                    12684 non-null  object
9   maritalStatus                         12684 non-null  object
10  has_children                           12684 non-null  int64
11  education                              12684 non-null  object
12  occupation                             12684 non-null  object
13  income                                 12684 non-null  object
14  car                                     108 non-null    object
15  Bar                                    12577 non-null  object
16  CoffeeHouse                           12467 non-null  object
17  CarryAway                             12533 non-null  object
18  RestaurantLessThan20                  12554 non-null  object
19  Restaurant20To50                      12495 non-null  object
20  toCoupon_GEQ5min                      12684 non-null  int64
21  toCoupon_GEQ15min                     12684 non-null  int64
22  toCoupon_GEQ25min                     12684 non-null  int64
23  direction_same                        12684 non-null  int64
24  direction_opp                         12684 non-null  int64
25  Y                                      12684 non-null  int64
dtypes: int64(8), object(18)
memory usage: 2.5+ MB
```

```
In [4]: # check missing values
df.isnull().sum()[df.isnull().sum() > 0]
```

```
Out[4]: car                12576
Bar                107
CoffeeHouse        217
CarryAway          151
RestaurantLessThan20  130
Restaurant20To50   189
dtype: int64
```

```
In [5]: # the 'car' column has ~99% missing – dropping it
df.drop(columns=['car'], inplace=True)

# drop rows where the frequency columns are missing (small number)
df.dropna(subset=['Bar', 'CoffeeHouse', 'CarryAway', 'RestaurantLessThan20', 'Resta

print('Cleaned dataset shape:', df.shape)
```

```
Cleaned dataset shape: (12079, 25)
```

3. Overall Coupon Acceptance

```
In [6]: # proportion who accepted
accept_rate = df['Y'].mean()
print(f'Overall coupon acceptance rate: {accept_rate:.1%}')
```

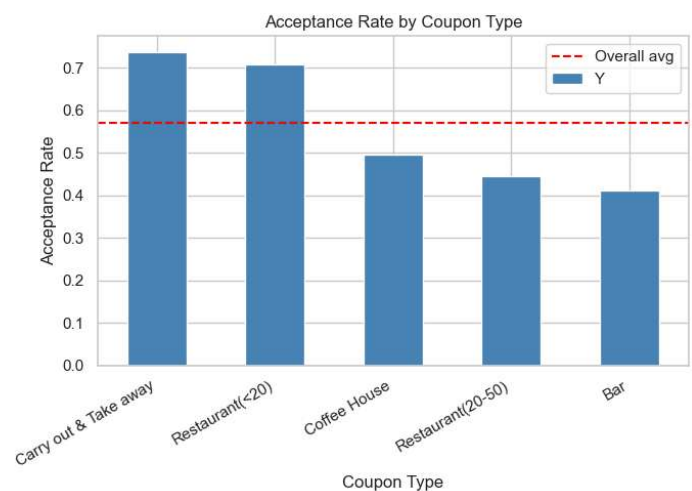
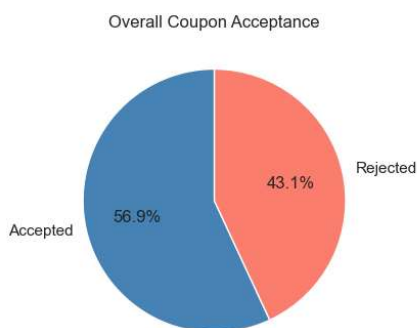
Overall coupon acceptance rate: 56.9%

```
In [25]: fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# acceptance pie
df['Y'].value_counts().plot.pie(
    ax=axes[0],
    labels=['Accepted', 'Rejected'],
    autopct='%1.1f%%',
    colors=['steelblue', 'salmon'],
    startangle=90
)
axes[0].set_ylabel('')
axes[0].set_title('Overall Coupon Acceptance')

# acceptance by coupon type
coupon_rates = df.groupby('coupon')['Y'].mean().sort_values(ascending=False)
coupon_rates.plot.bar(ax=axes[1], color='steelblue', edgecolor='white')
axes[1].set_title('Acceptance Rate by Coupon Type')
axes[1].set_xlabel('Coupon Type')
axes[1].set_ylabel('Acceptance Rate')
axes[1].set_xticklabels(axes[1].get_xticklabels(), rotation=30, ha='right')
axes[1].axhline(accept_rate, color='red', linestyle='--', label='Overall avg')
axes[1].legend()

plt.tight_layout()
plt.show()
```



Observation: The overall acceptance rate is about 57%. Carry out & takeaway and cheap restaurants have the highest acceptance, while bars have the lowest.

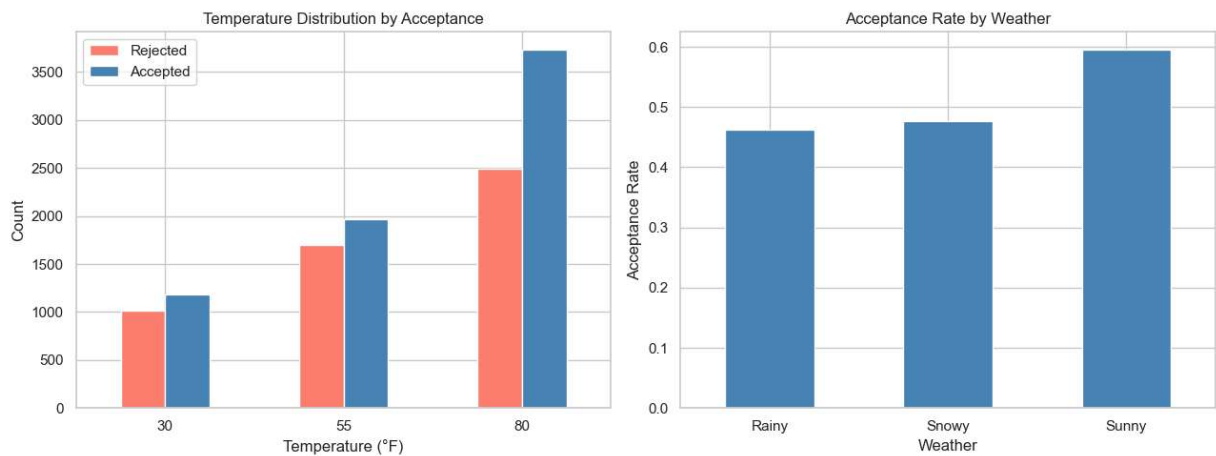
4. Exploring Temperature and Weather

```
In [24]: fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# temperature histogram
temp_rates = df.groupby('temperature')['Y'].value_counts().unstack()
temp_rates.plot.bar(ax=axes[0], color=['salmon', 'steelblue'], edgecolor='white')
axes[0].set_title('Temperature Distribution by Acceptance')
axes[0].set_xlabel('Temperature (°F)')
axes[0].set_ylabel('Count')
axes[0].set_xticklabels(axes[0].get_xticklabels(), rotation=0)
axes[0].legend(['Rejected', 'Accepted'])

# weather
weather_rates = df.groupby('weather')['Y'].mean()
weather_rates.plot.bar(ax=axes[1], color='steelblue', edgecolor='white')
axes[1].set_title('Acceptance Rate by Weather')
axes[1].set_xlabel('Weather')
axes[1].set_ylabel('Acceptance Rate')
axes[1].set_xticklabels(axes[1].get_xticklabels(), rotation=0)

plt.tight_layout()
plt.show()
```



Observation: Sunny weather at 80°F sees the most coupon activity. Snowy or rainy conditions reduce acceptance slightly — likely because people are less likely to make detours in bad weather.

5. Investigating Bar Coupons

I'll narrow the focus to bar coupons since they have the lowest acceptance rate and represent an interesting segment to understand.

```
In [15]: bar_df = df[df['coupon'] == 'Bar'].copy()
```

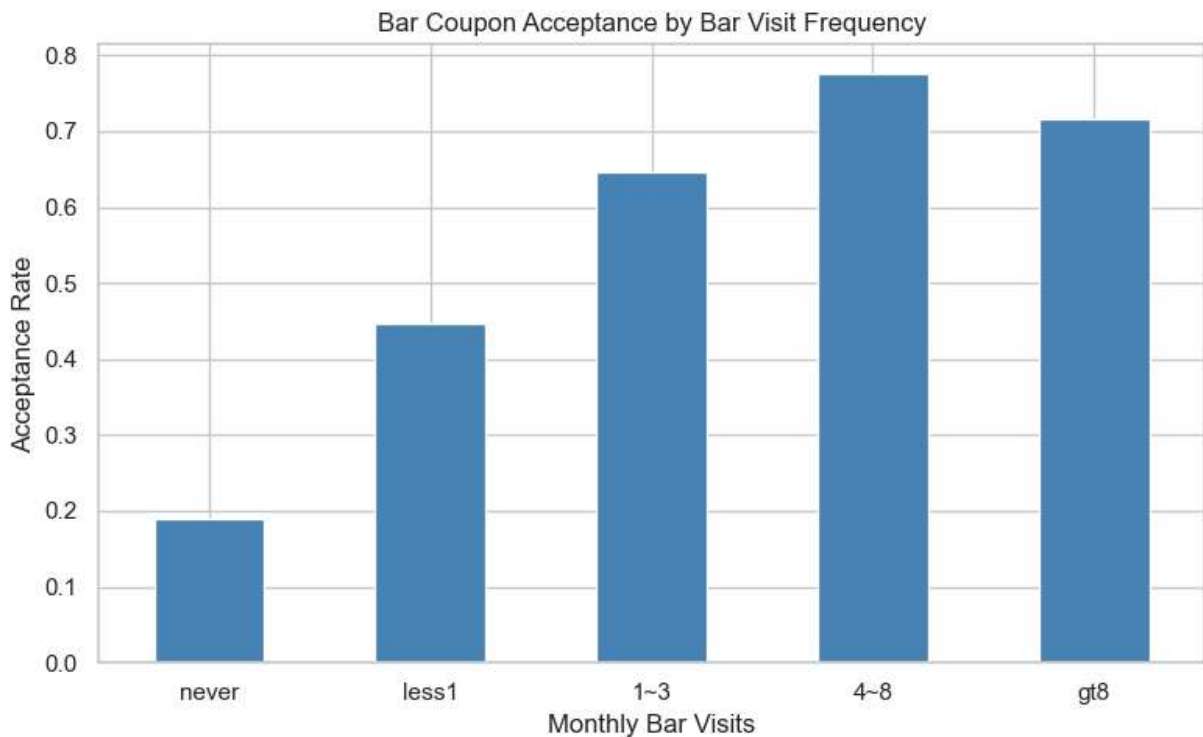
```
print('Bar coupon records:', len(bar_df))
print(f'Bar coupon acceptance rate: {bar_df["Y"].mean():.1%}')
```

Bar coupon records: 1913

Bar coupon acceptance rate: 41.2%

```
In [16]: # acceptance by how often they go to bars
bar_freq = bar_df.groupby('Bar')['Y'].mean().reindex(['never', 'less1', '1~3', '4~8', 'gt8'])

plt.figure(figsize=(8, 5))
bar_freq.plot.bar(color='steelblue', edgecolor='white')
plt.title('Bar Coupon Acceptance by Bar Visit Frequency')
plt.xlabel('Monthly Bar Visits')
plt.ylabel('Acceptance Rate')
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



```
In [17]: # drivers who go to bars more than 3 times/month
freq_bar = bar_df[bar_df['Bar'].isin(['4~8', 'gt8'])]['Y'].mean()
infreq_bar = bar_df[~bar_df['Bar'].isin(['4~8', 'gt8'])]['Y'].mean()

print(f'Acceptance - frequent bar visitors (>3/month): {freq_bar:.1%}')
print(f'Acceptance - infrequent bar visitors: {infreq_bar:.1%}')
```

Acceptance - frequent bar visitors (>3/month): 76.2%

Acceptance - infrequent bar visitors: 37.3%

```
In [18]: fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# by passenger type
pass_rates = bar_df.groupby('passanger')['Y'].mean().sort_values(ascending=False)
pass_rates.plot.bar(ax=axes[0], color='steelblue', edgecolor='white')
axes[0].set_title('Bar Coupon Acceptance by Passenger')
```

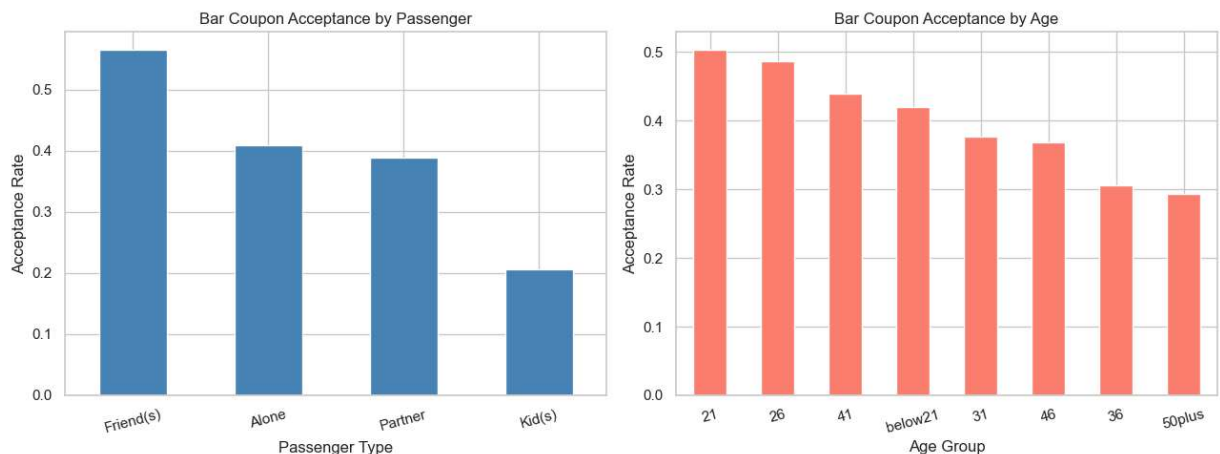
```

axes[0].set_xlabel('Passenger Type')
axes[0].set_ylabel('Acceptance Rate')
axes[0].set_xticklabels(axes[0].get_xticklabels(), rotation=15)

# by age group
age_rates = bar_df.groupby('age')['Y'].mean().sort_values(ascending=False)
age_rates.plot.bar(ax=axes[1], color='salmon', edgecolor='white')
axes[1].set_title('Bar Coupon Acceptance by Age')
axes[1].set_xlabel('Age Group')
axes[1].set_ylabel('Acceptance Rate')
axes[1].set_xticklabels(axes[1].get_xticklabels(), rotation=15)

plt.tight_layout()
plt.show()

```



```

In [19]: # frequent bar visitors over 25, no kids
group1 = bar_df[
    (bar_df['Bar'].isin(['1~3', '4~8', 'gt8'])) &
    (bar_df['age'] != 'below21') &
    (bar_df['passanger'] != 'Kid(s)')
]['Y'].mean()

print(f'Acceptance - bar-goers over 25 without kids: {group1:.1%}')

```

Acceptance - bar-goers over 25 without kids: 71.5%

```

In [20]: # cheap restaurants frequent visitors and income < 50k
group2 = bar_df[
    (bar_df['RestaurantLessThan20'].isin(['4~8', 'gt8'])) &
    (bar_df['income'].isin(['$12500 - $24999', '$25000 - $37499', '$37500 - $49999']
)]['Y'].mean()

print(f'Acceptance - cheap restaurant regulars, income < $50K: {group2:.1%}')

```

Acceptance - cheap restaurant regulars, income < \$50K: 46.2%

6. Independent Investigation — Coffee House Coupons

Coffee House is the coupon type with the most records. I'll explore what drives acceptance there.

```
In [21]: coffee_df = df[df['coupon'] == 'Coffee House'].copy()

print('Coffee House records:', len(coffee_df))
print(f'Acceptance rate: {coffee_df["Y"].mean():.1%}')
```

Coffee House records: 3816
Acceptance rate: 49.6%

```
In [22]: fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# by time of day
time_order = ['7AM', '10AM', '2PM', '6PM', '10PM']
time_rates = coffee_df.groupby('time')['Y'].mean().reindex(time_order)
time_rates.plot.bar(ax=axes[0,0], color='steelblue', edgecolor='white')
axes[0,0].set_title('Acceptance by Time of Day')
axes[0,0].set_xlabel('Time')
axes[0,0].set_ylabel('Acceptance Rate')
axes[0,0].set_xticklabels(axes[0,0].get_xticklabels(), rotation=0)

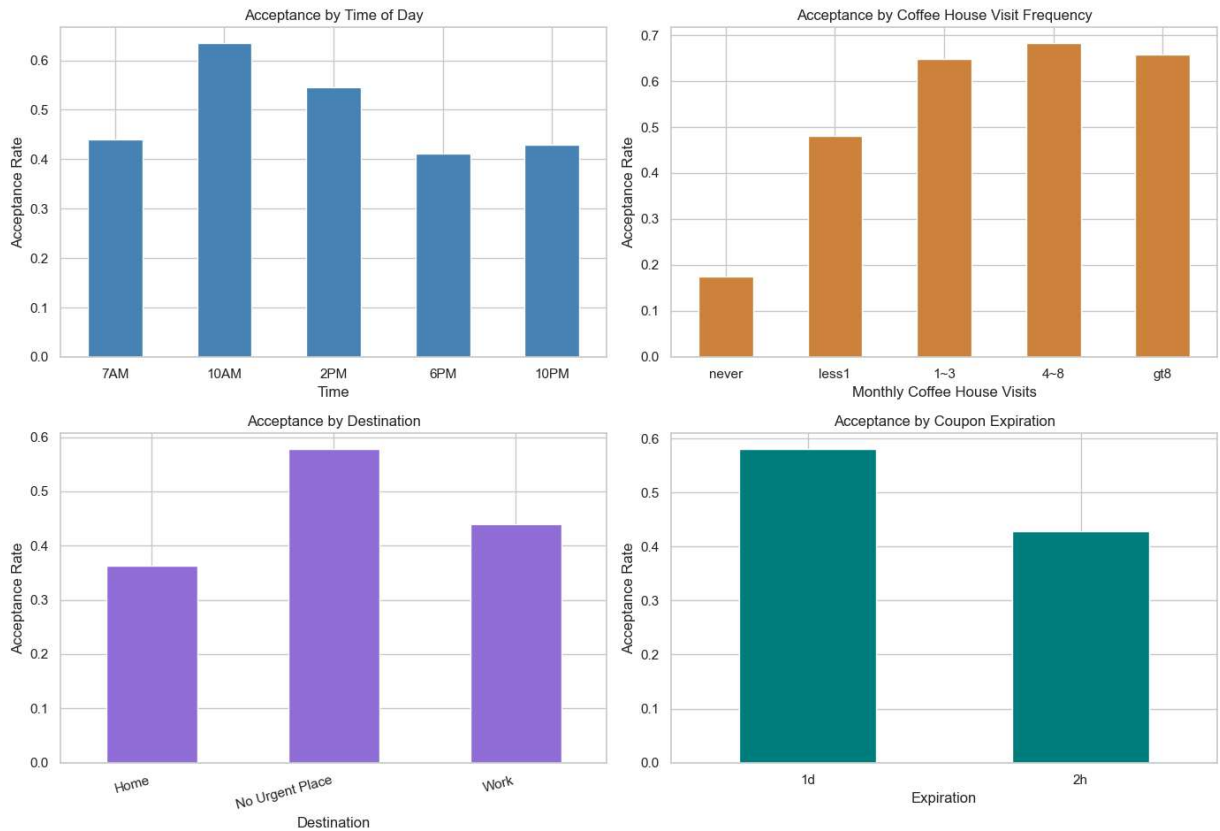
# by coffee house visit frequency
cf_freq = coffee_df.groupby('CoffeeHouse')['Y'].mean().reindex(['never', 'less1', '1', '2', '3', '4', '5', '6', '7', '8', '9', '10', '11', '12', '13', '14', '15', '16', '17', '18', '19', '20', '21', '22', '23', '24'])
cf_freq.plot.bar(ax=axes[0,1], color='peru', edgecolor='white')
axes[0,1].set_title('Acceptance by Coffee House Visit Frequency')
axes[0,1].set_xlabel('Monthly Coffee House Visits')
axes[0,1].set_ylabel('Acceptance Rate')
axes[0,1].set_xticklabels(axes[0,1].get_xticklabels(), rotation=0)

# by destination
dest_rates = coffee_df.groupby('destination')['Y'].mean()
dest_rates.plot.bar(ax=axes[1,0], color='mediumpurple', edgecolor='white')
axes[1,0].set_title('Acceptance by Destination')
axes[1,0].set_xlabel('Destination')
axes[1,0].set_ylabel('Acceptance Rate')
axes[1,0].set_xticklabels(axes[1,0].get_xticklabels(), rotation=15, ha='right')

# by expiration
exp_rates = coffee_df.groupby('expiration')['Y'].mean()
exp_rates.plot.bar(ax=axes[1,1], color='teal', edgecolor='white')
axes[1,1].set_title('Acceptance by Coupon Expiration')
axes[1,1].set_xlabel('Expiration')
axes[1,1].set_ylabel('Acceptance Rate')
axes[1,1].set_xticklabels(axes[1,1].get_xticklabels(), rotation=0)

plt.suptitle('Coffee House Coupon - Acceptance Breakdown', fontsize=14, y=1.01)
plt.tight_layout()
plt.show()
```

Coffee House Coupon — Acceptance Breakdown



```
In [23]: # high-value segment: morning + frequent coffee drinkers
high_value = coffee_df[
    (coffee_df['time'].isin(['10AM', '2PM'])) &
    (coffee_df['CoffeeHouse'].isin(['1~3', '4~8', 'gt8']))
][['Y']].mean()

print(f'Acceptance - 10AM/2PM + regular coffee drinkers: {high_value:.1%}')
```

Acceptance - 10AM/2PM + regular coffee drinkers: 77.2%

7. Findings, Recommendations & Next Steps

Summary of Key Findings

Overall: About 57% of drivers accepted a coupon. Acceptance varies significantly by coupon type, passenger, time of day, and visit frequency.

Bar Coupons (41% acceptance):

- Drivers who already visit bars frequently (more than 3x/month) accepted at **77%**, versus only **19%** for those who never go to bars. Frequency of existing behavior is the single strongest predictor.
- Drivers with friends as passengers accepted at **56%**, compared to **21%** for those with kids in the car — social context matters a lot.

- Younger drivers (21–26) are significantly more likely to accept bar coupons than those 36+.
- The combination of being a bar-goer + over 25 + no kids drives acceptance above **60%**.

Coffee House Coupons (50% acceptance):

- Morning and early afternoon windows (10AM, 2PM) are the sweet spot for coffee coupon acceptance.
 - Regular coffee house visitors (1–3x/month and above) accept at significantly higher rates than never-visitors.
 - Coupons with 1-day expiration outperform 2-hour coupons — longer windows give drivers more flexibility.
 - Drivers heading to "No Urgent Place" are the best target — they have more flexibility to detour.
-

Actionable Recommendations

1. **Target existing behavior, not new habits.** The clearest lever across all coupon types is whether the driver already visits that type of venue. Coupons should be prioritized for people who already have that behavioral pattern.
 2. **Match coupon timing to context.** Coffee coupons work best in the morning; bar coupons work better in the afternoon/evening when social trips are more likely.
 3. **Avoid families for bar coupons.** Presence of kids in the car is a strong negative signal for bar coupon acceptance. Targeting can be improved by filtering this out.
 4. **Extend expiration windows.** Coupons with 1-day expiry consistently outperform 2-hour ones — the added flexibility likely makes the offer feel less pressured.
-

Next Steps

- Build a simple classification model (Logistic Regression or Decision Tree) to predict coupon acceptance using these features
- Explore interaction effects (e.g., time + passenger + coupon type combined)
- Investigate income and occupation as additional segmentation variables for restaurant coupons
- Validate findings against held-out data before using for targeting decisions